

Design Report BGR

Maehashi

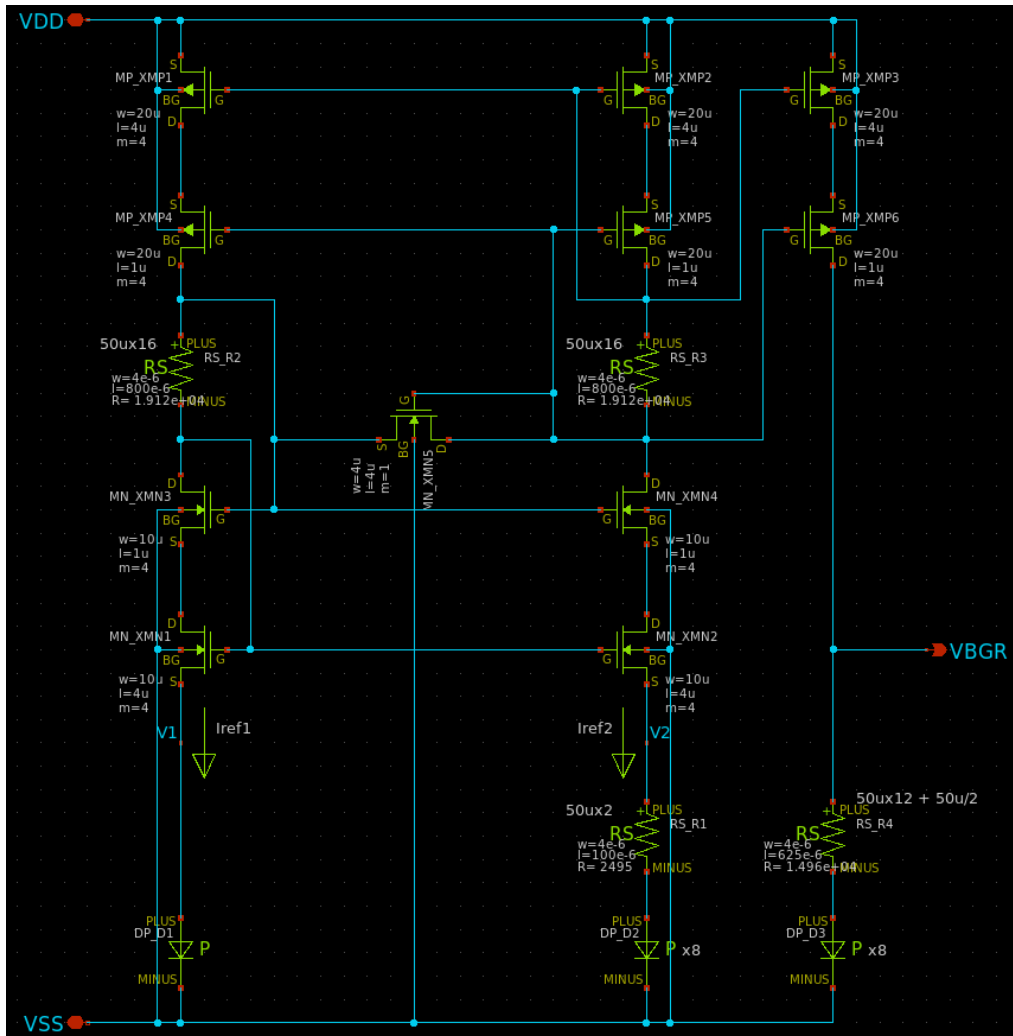
Overview

- Process TR10 (OpenSUSI TR-1um)
- Tools Circuit Design: Xschem, Layout Design: Klayout
- Target Spec. VDD 5.0V $\pm$ 0.5V
 Temp. 0°C ~ 80°C
 Process MOS Vth $\pm$ 0.1V, Poly-Res. $\pm$ 15%

Result		bgr_a	bgr_b
Vout	Typ (TT/5.0V/40°C)	1.094 V	2.492 V
	Worst @ 0°C	1.111 V (+1.6%)	2.509 V (+0.7%)
	Worst @ 80°C	1.087 V (-0.6%)	2.475 V (-0.7%)
Iout	Typ (TT/5.0V/40°C)	-	46.2 uA
	Worst @ 0°C	-	40.0 uA (-15.4%)
	Worst @ 80°C	-	54.7 uA (+18.4%)

bgr_a

- Schematic



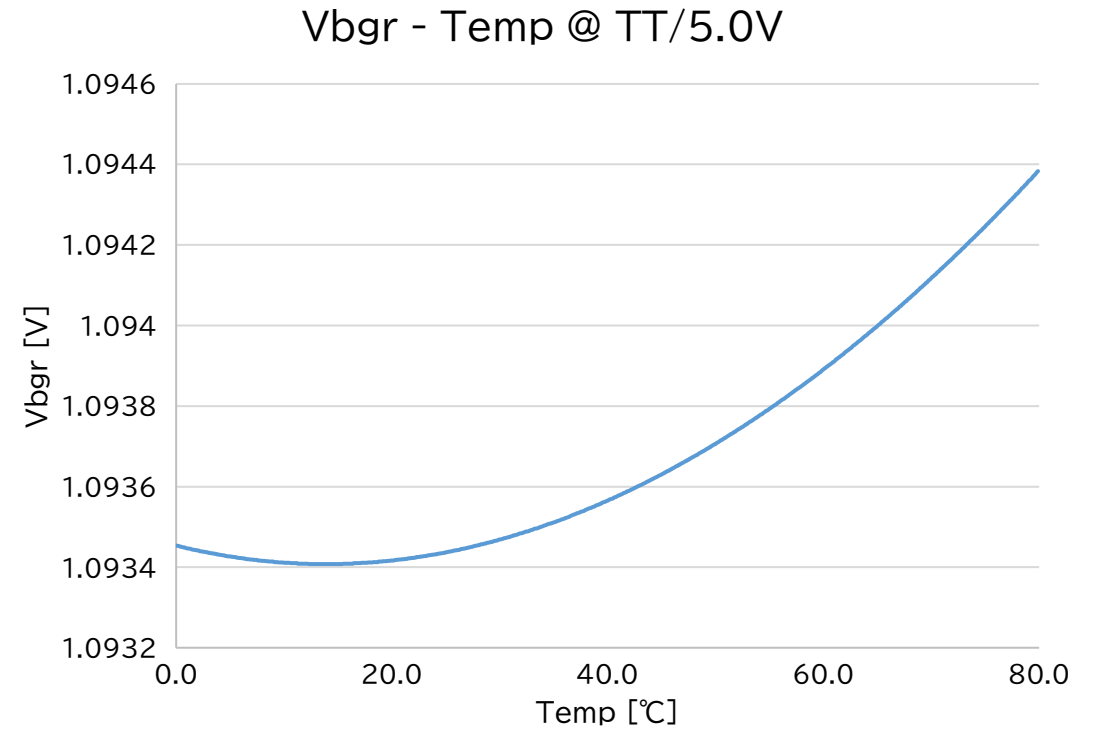
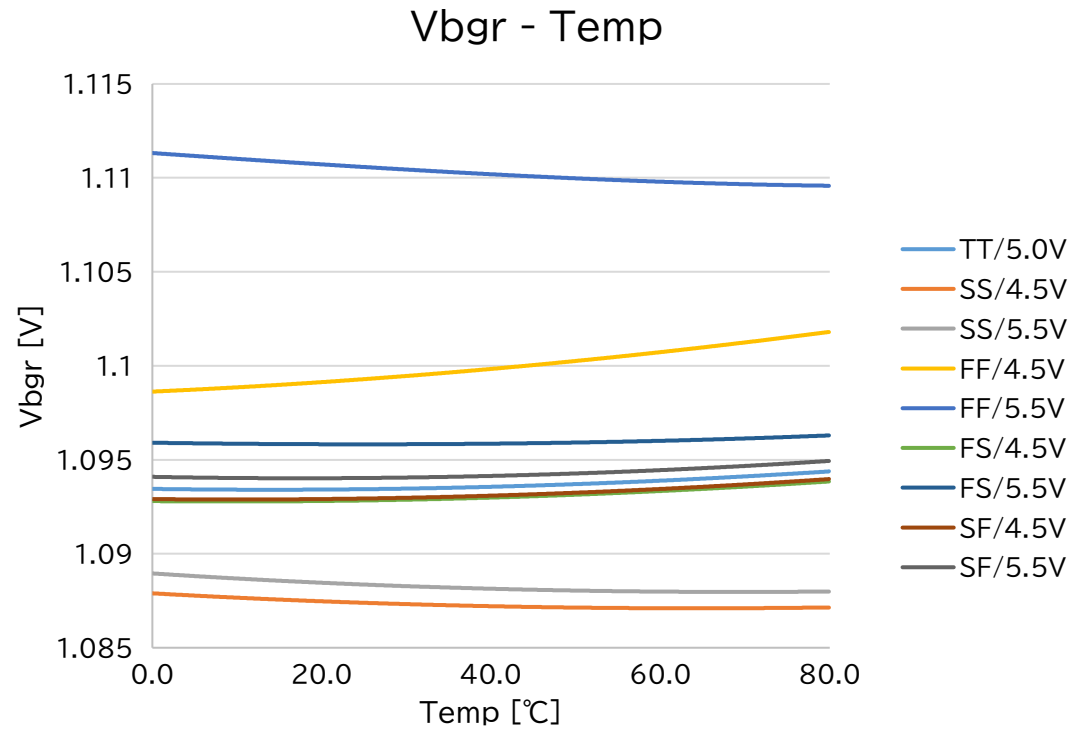
Architecture^[*]

- Iptat Gen. Circuit + Vctat Gen. Diode
- Self-biased Cascode
- Startup Transistor

[*] B. Razavi, "Design of Analog CMOS Integrated Circuit"

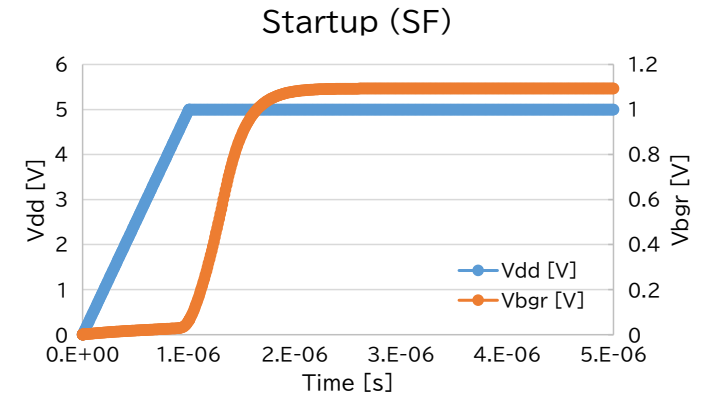
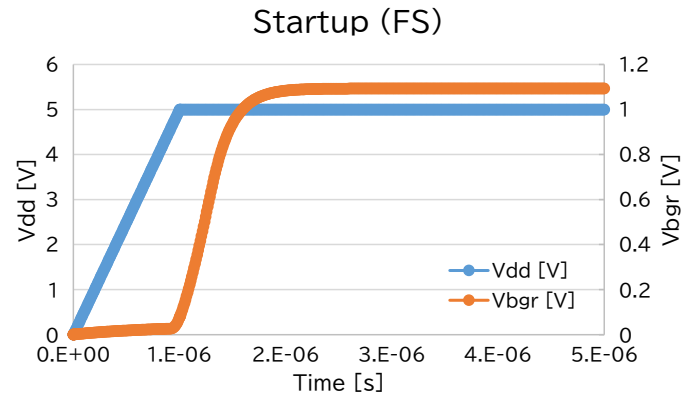
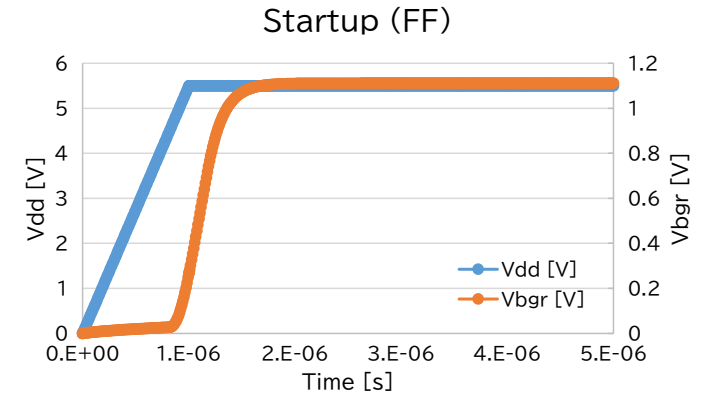
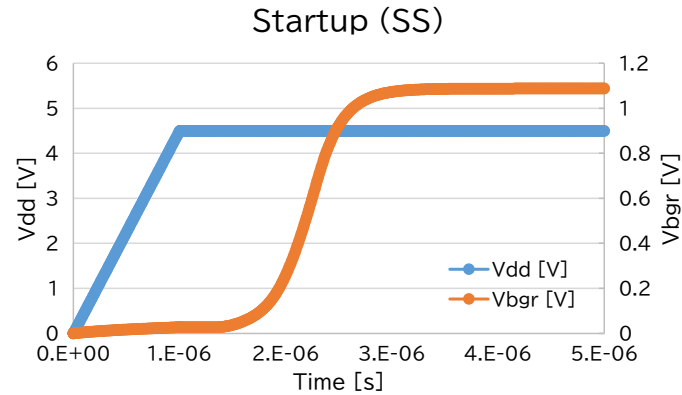
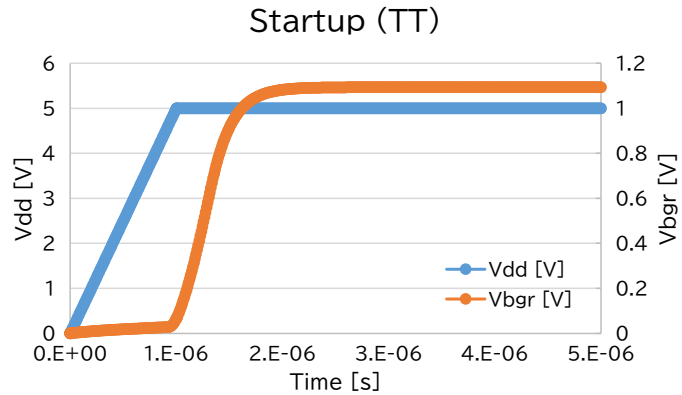
bgr_a

- DC Characteristics



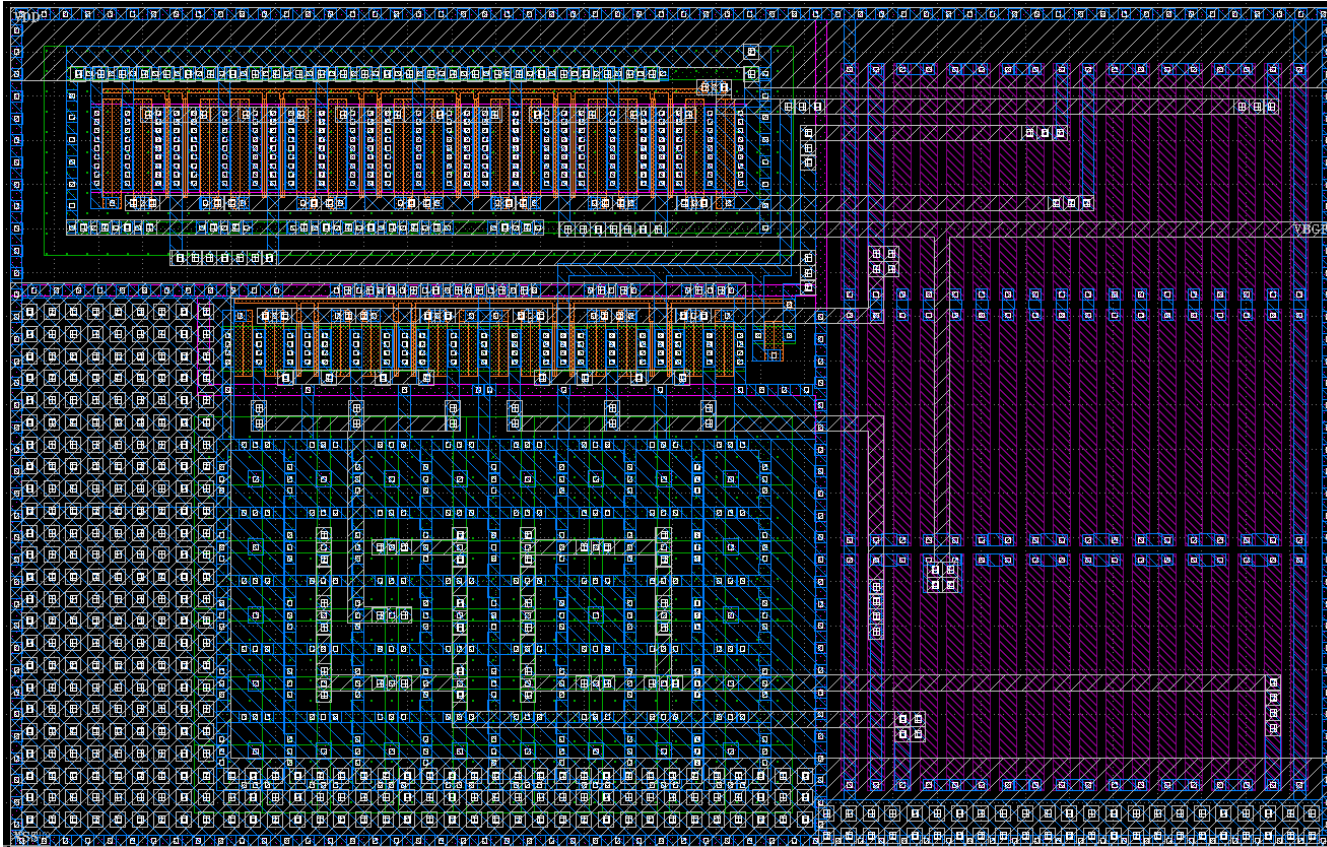
bgr_a

- Tran Characteristics (Startup)



bgr_a

- Layout

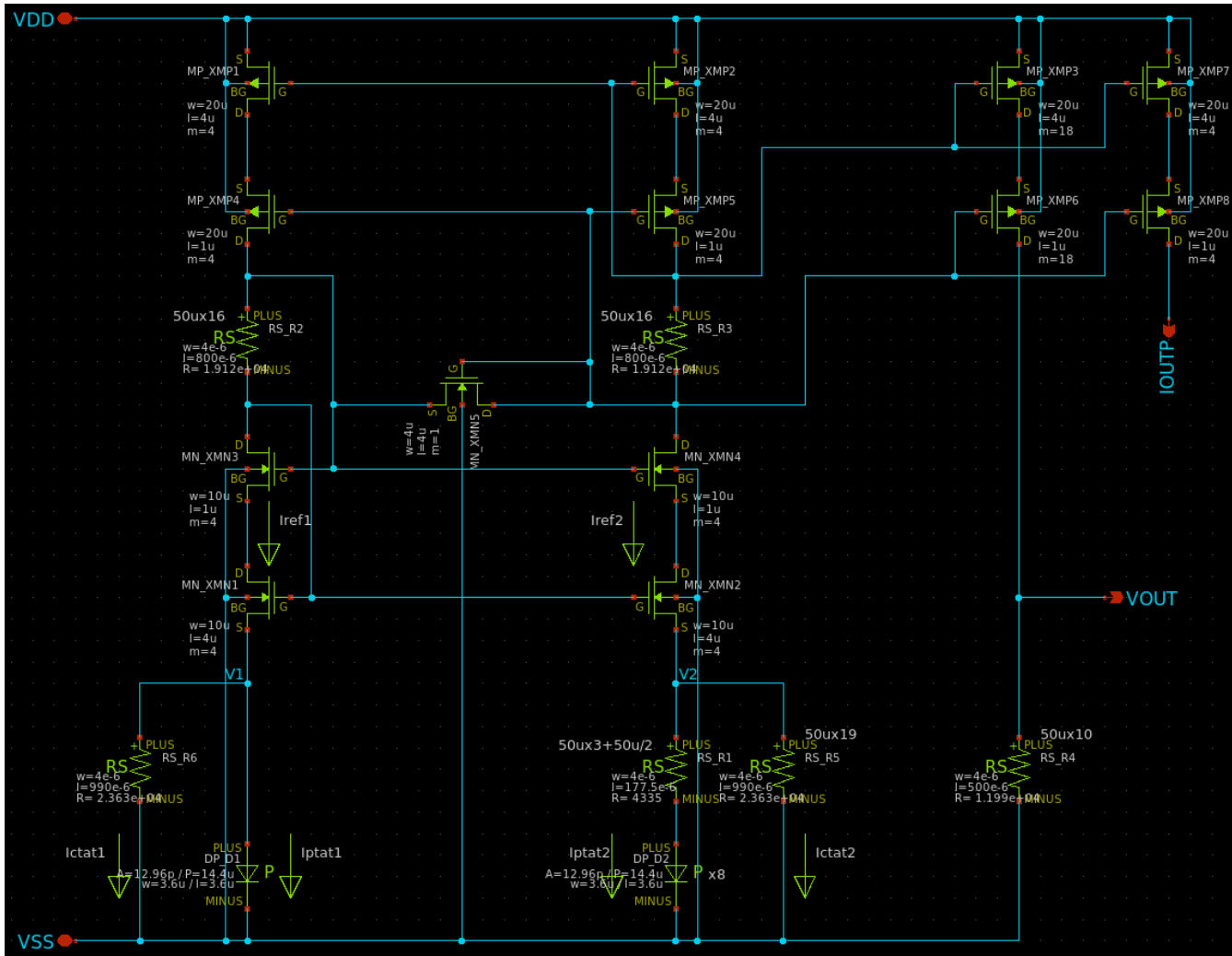


Features

- 300um x 190um Area usage
- Employing poly resistor for stability against temperature variation

bgr_b

- Schematic



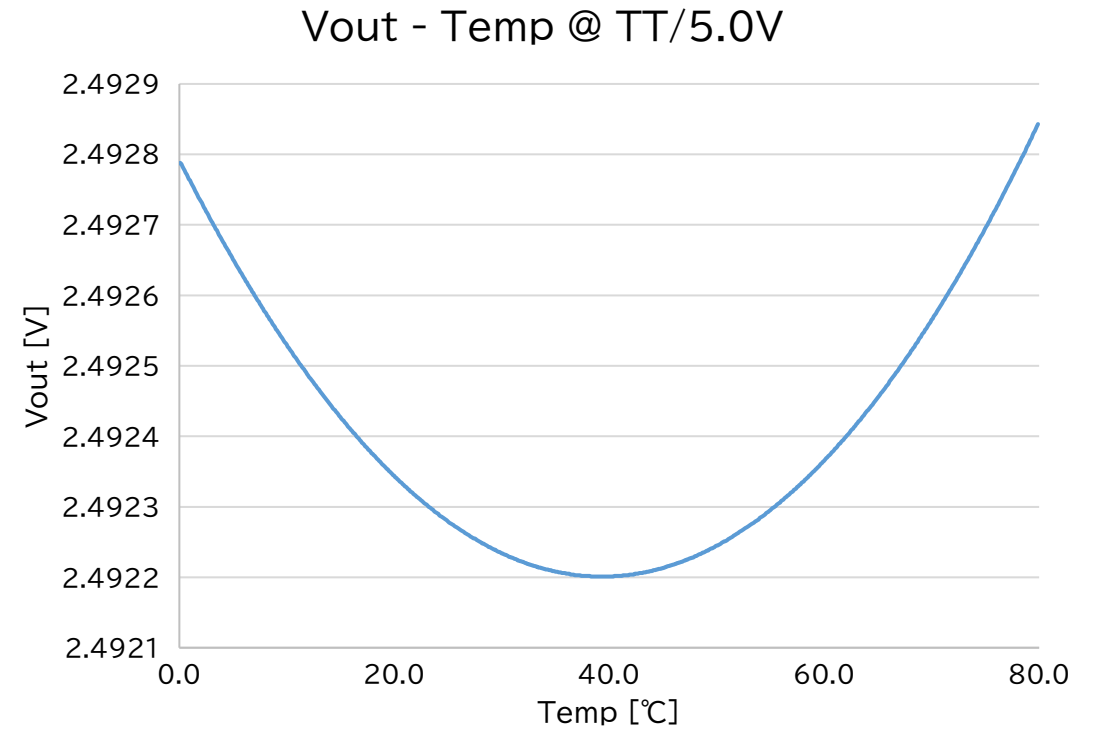
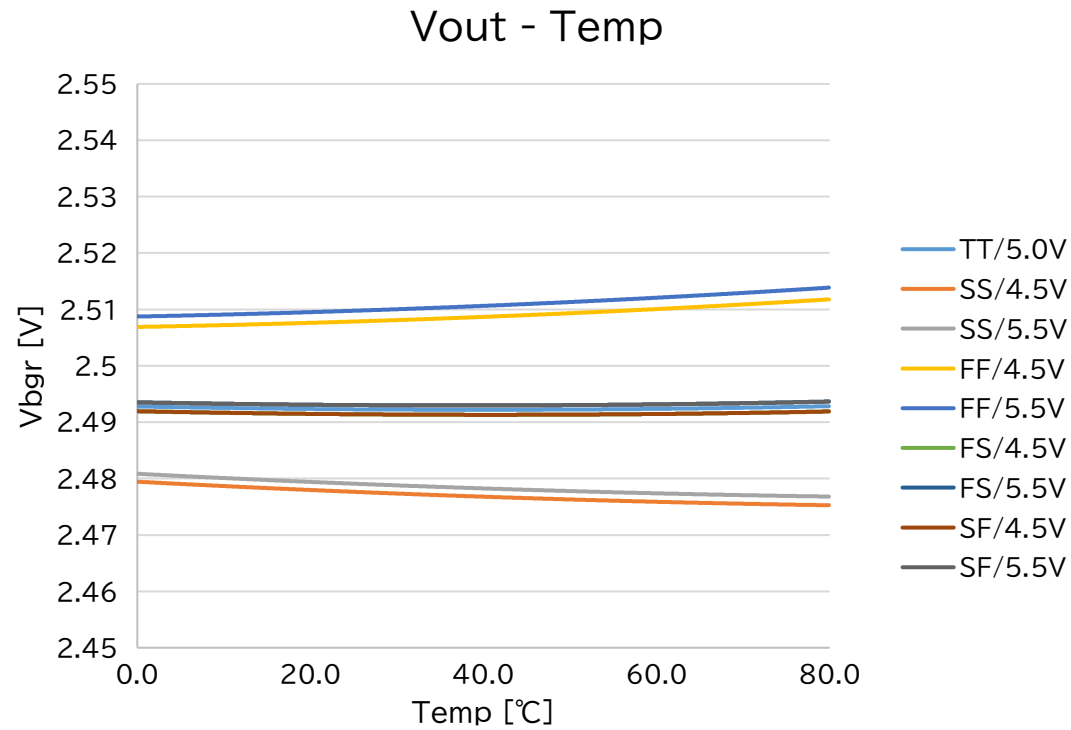
Architecture^[*]

- Iconst Gen. Circuit + Resistor
- Self-biased Cascode
- Startup Transistor

[*] B. Razavi, "Design of Analog CMOS Integrated Circuit"

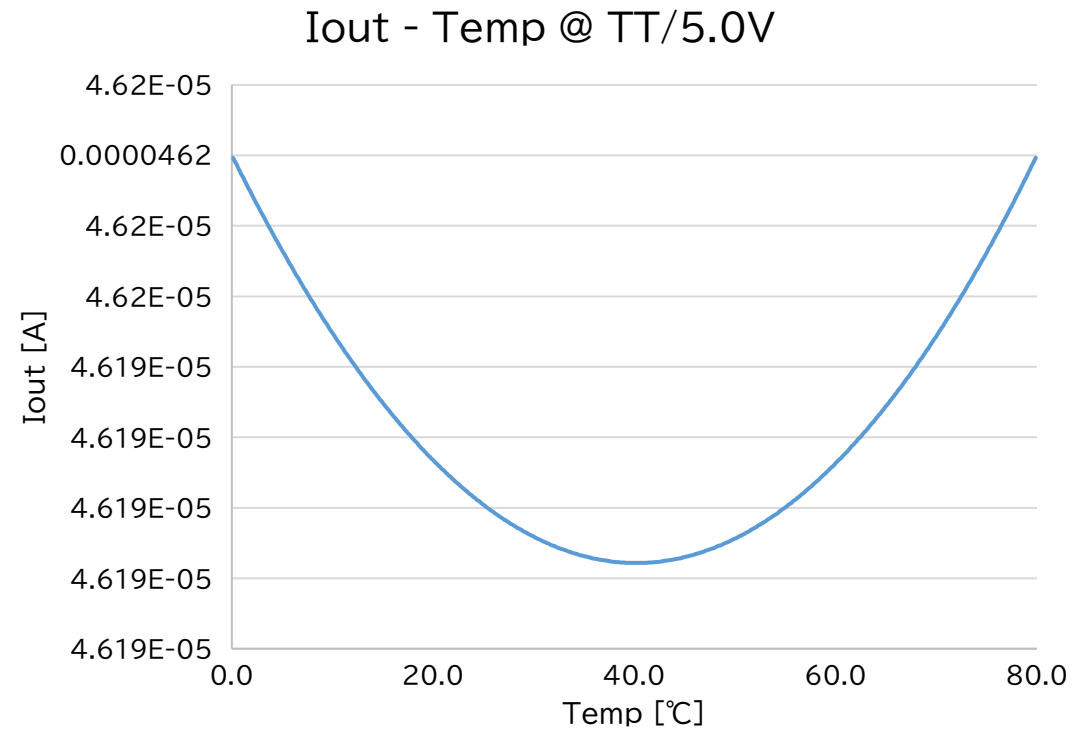
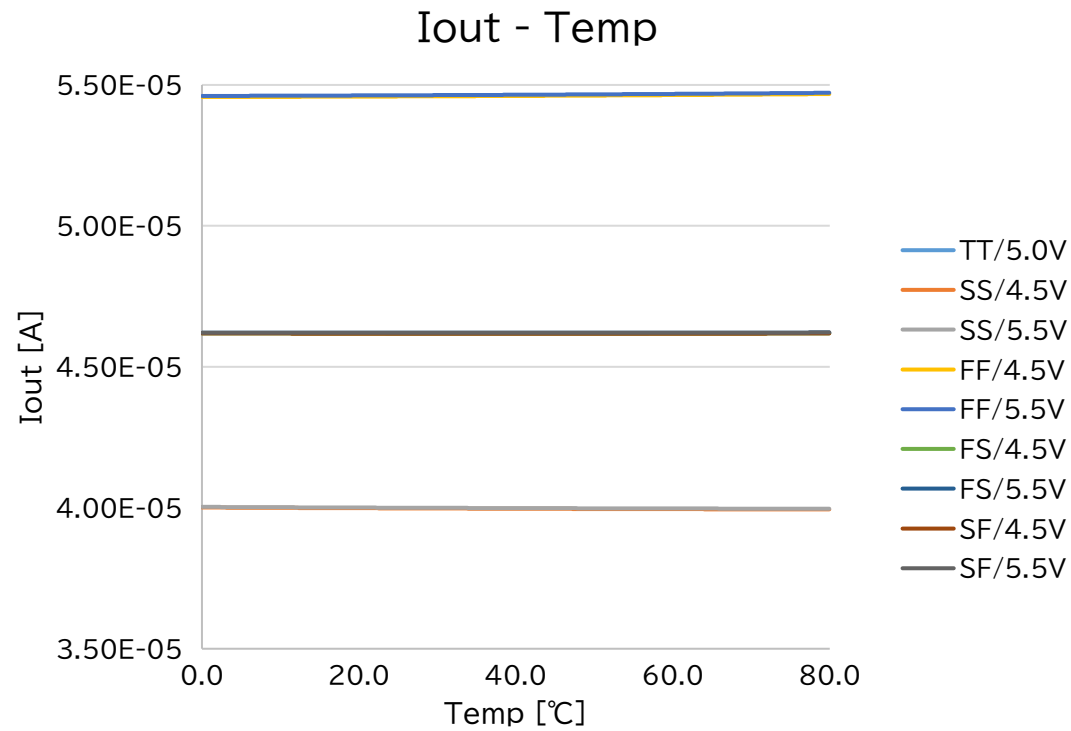
bgr_b

- DC Characteristics @ Vout



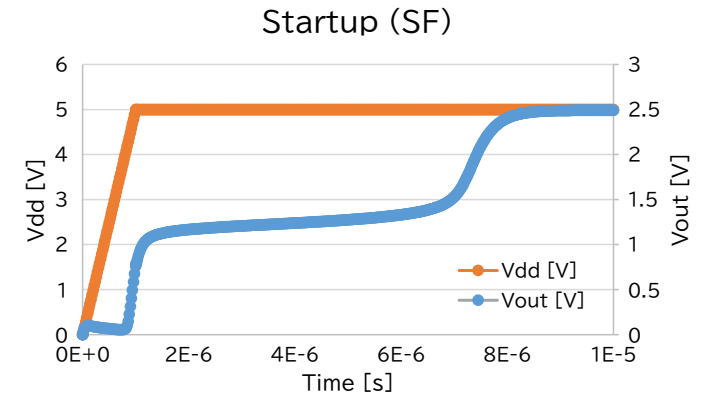
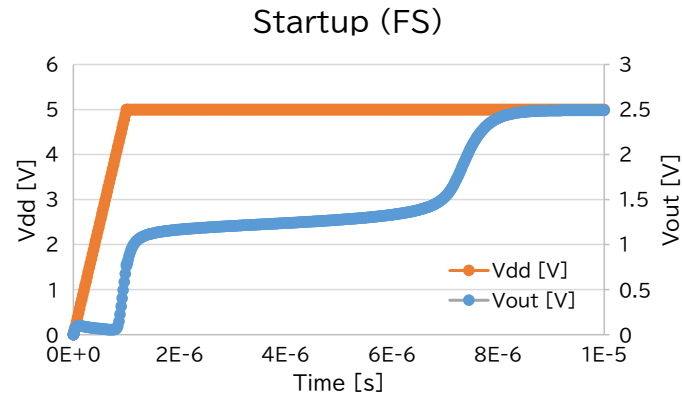
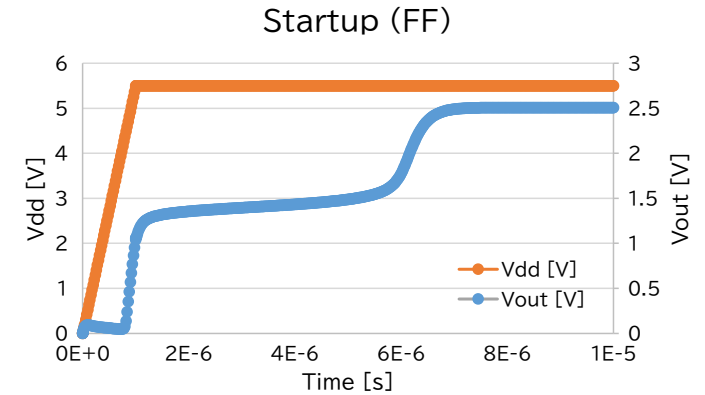
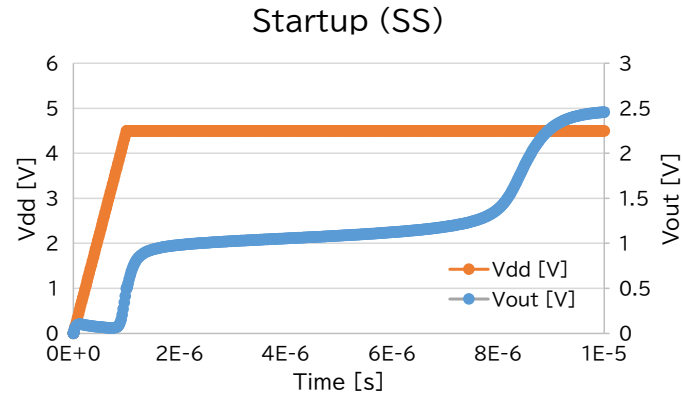
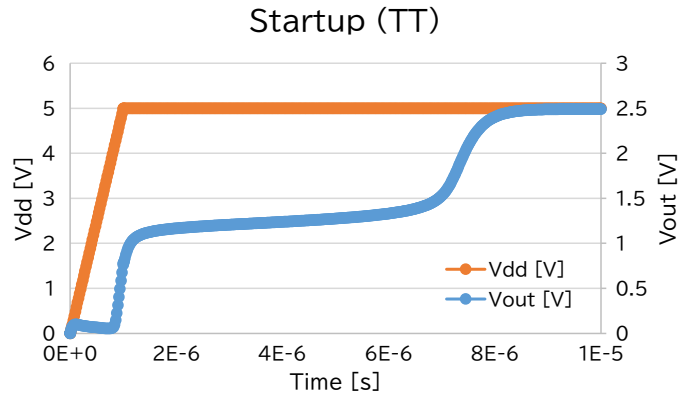
bgr_b

- DC Characteristics @ Iout



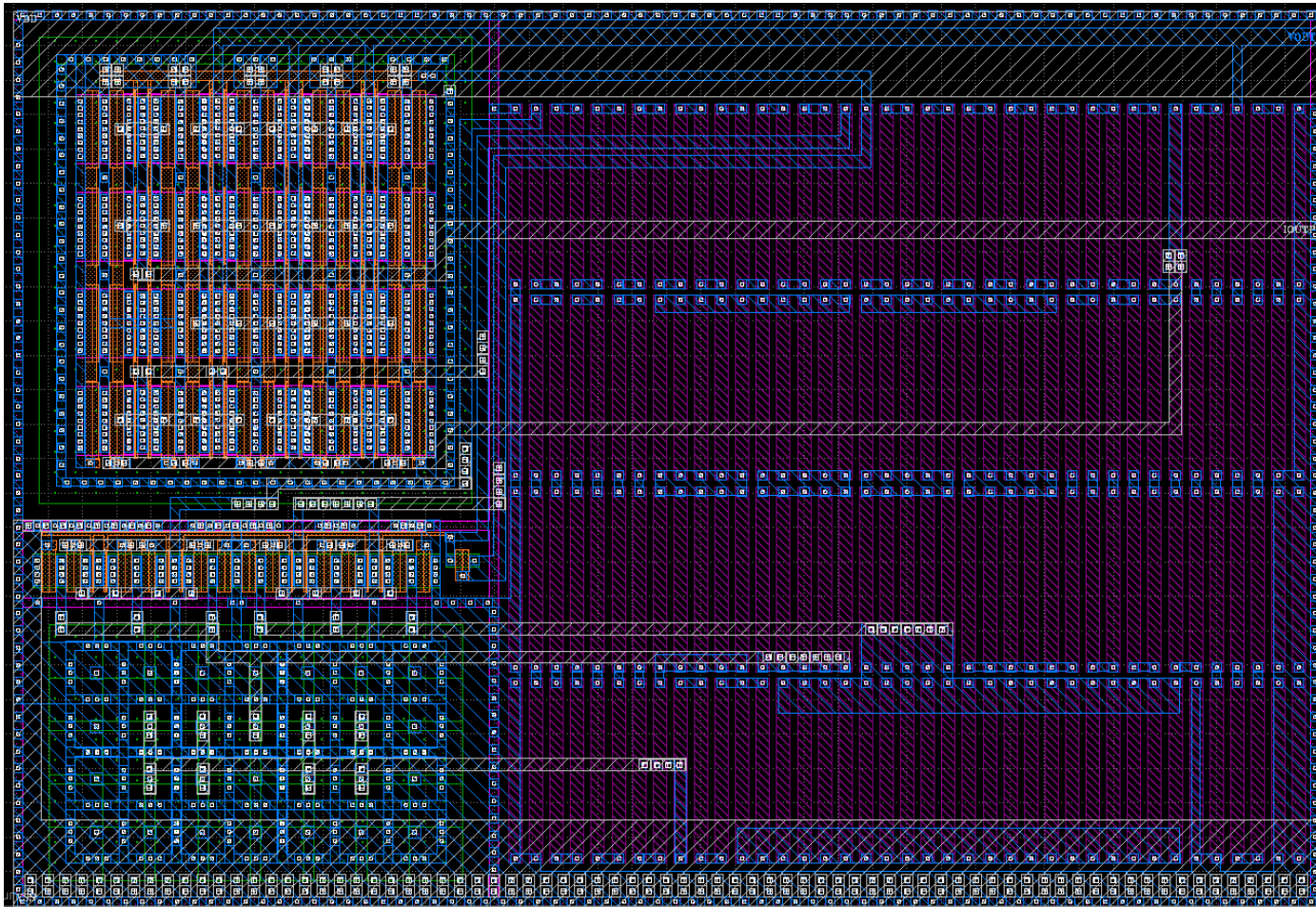
bgr_b

- Tran Characteristics (Startup)



bgr_b

- Layout



Features

- 380um x 260um Area usage
- Employing poly resistor for stability against temperature variation

bgr_b

- Schematic w/ dummy elements (for LVS)

